Sahaj Bhadja

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Summary

Backend engineer with 2+ years building and operating **cloud-native distributed systems** with strong focus on **reliability**, **performance**, **observability-driven debugging**, and production ownership across **Python**, **Java**, **SQL**, and **Kubernetes-adjacent** tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built and operated **multi-tenant backend services** on **Azure SQL/PostgreSQL**, supporting 500+ users and reducing latency from **1.2s to 350ms** through query/index tuning.
- Implemented **asynchronous validation/release workflows** for 5K+ monthly jobs with reconciliation checks, improving **production reliability**, traceability, and incident recovery.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed and shipped backend-heavy features in a **Flask + React** system handling **10K+ daily requests**, with modular APIs and high-throughput request paths.
- Added **concurrent execution** and **CI/CD controls** via GitHub Actions, reducing turnaround time by **60%** while preserving stable runtime behavior at peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **simulation pipelines** in Python processing **100K+ trajectory logs**, improving long-horizon planning performance by **18%** with repeatable experiment runs.
- Produced structured validation reports and debug traces, improving **observability of model/system behavior** and faster cross-team issue triage.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **microservice-style API/ETL workflows** using OOP, AWS Glue, and S3, improving response times by **30%** and pipeline efficiency by **68%**.
- Maintained **20+ unit/integration test suites** (Jenkins, PyTest, Mockito) across 3 OS environments, strengthening release quality and on-call troubleshooting confidence.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **REST API** and **WebSocket** services for 8+ workflows and 2M+ synchronized records with robust operational logging and diagnostics.
- Automated deployment/testing with **Jenkins + Docker**, reducing deployment time by **50%** and improving service rollout reliability.

Skills

Languages: **Python**, **Java**, SQL, JavaScript/TypeScript, C++
Distributed Systems: microservices architecture, async workflows, reliability engineering, performance tuning
Cloud & Infra: Kubernetes familiarity, AWS (S3, Glue, RDS), Azure SQL, Docker, CI/CD, infrastructure-as-code familiarity
Observability: metrics/logs analysis, incident debugging, system health monitoring, production support
Data Platforms: PostgreSQL, MySQL, SQL Server, MongoDB, high-volume query optimization
Engineering: code reviews, design discussions, Agile/Scrum, on-call readiness, technical documentation

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built **semantic retrieval services** over 50K+ documents with efficient indexing/query paths, enabling low-latency summarization and production-style scaling patterns.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built a **multi-database backend platform** supporting 10+ databases with schema-aware query generation, robust execution, and maintainable service boundaries.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Implemented high-scale transaction workflows supporting **50K+ users** and **1M+ transactions** with real-time updates, secure APIs, and cloud deployment resilience.